
7-Day HTML, CSS & React Roadmap

Day 1 – HTML + Basic CSS

Topics Covered:

- HTML5 structure (doctype, meta, head, body)
- Tags: headings, links, lists, images, forms, tables
- CSS Basics: selectors, colors, spacing, fonts, borders
- Internal and External CSS

Practice Projects:

- Resume Page
 - Registration Form (styled)
 - Weekly Timetable using HTML tables
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Day 2 – Material UI (MUI)

Topics Covered:

- Introduction to MUI and installation
- Theme provider and customization
- Common components: Button, Typography, Card, AppBar, Grid, Modal
- Icons and responsive layout using MUI Grid system

Practice Projects:

- MUI-based Login Page
 - Dashboard layout with AppBar and Drawer
 - Product Card using MUI Grid and custom theme
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Day 3 – React: Setup & Fundamentals

Topics Covered:

- Setting up with create-react-app or Vite
- JSX syntax and rules
- Functional components and props
- useState hook and event handling

Practice Projects:

- Greeting component with props
 - Counter App with increment/decrement
 - Toggle visibility feature
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Day 4 – Lists, Conditional Rendering & UI Components**Topics Covered:**

- Rendering lists with `.map()`
- Proper usage of the `key` prop
- Conditional rendering (if, ternary, logical `&&`)
- Component structure and organization

Practice Projects:

- To-do App with delete functionality
 - FAQ Accordion component
 - Reusable Button and Card components
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Day 5 – React Forms & Controlled Inputs**Topics Covered:**

- Controlled components using `value` and `onChange`
- Basic form validation
- Handling form submissions
- `useState` with object-based form state

Practice Projects:

- Signup form with validation
 - Search filter for a list of items
 - Feedback form with success message
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Day 6 – `useEffect` & Common React Hooks**Topics Covered:**

- `useEffect` for data fetching and side effects
- Conditional triggers for `useEffect`

- useState + useEffect integration
- Additional hooks:
 - useRef for DOM access and persistent values
 - useMemo for optimizing expensive calculations
 - useCallback to memoize functions

Practice Projects:

- Fetch and display user list from API
 - Handle loading and error states
 - Real-time character counter with useEffect
 - Scroll-to-top button using useRef
 - Memoized calculator (e.g., factorial)
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Day 7 – Component Composition & Mini Project

Topics Covered:

- Smart vs dumb components
- Prop drilling and state management
- Splitting UI into reusable components
- Lifting state up to parent components

Mini Project: E-Commerce UI

- Product List (API or hardcoded)
- Cart component (add/remove items)
- Filter system (by category or price)
- Header with total cart count